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A Fuzzy LARG Index for Assessing the Lean, Agile, Resilience, and Green Paradigms in Industrial Companies

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Abstract: Today's sharp competition has forced organizations to adopt effective improvement paradigms, including lean, agile, green, and resilience (LARG). However, an assessment tool is necessary to monitor the progress of *LARG* adoption and evaluate its effectiveness. This research, therefore, developed an index for assessing the effectiveness of *LARG* paradigms by evaluating their principles and practices with experts' fuzzy evaluations. Initially, thorough research on *LARG* paradigms was conducted to determine the paradigm principles and their measures and prepare a comprehensive *LARG* survey. The Delphi method with four experts was used to rate item measures of *LARG* based on a five-point Likert scale. The principles and measures of each paradigm were represented by triangular fuzzy membership functions. Then, defuzzified values were obtained for each principle and set as inputs in the fuzzy inference system (*FIS*) to obtain a crisp value for each paradigm. Next, the defuzzified values of lean, agile, and green (*LAG*) were input in the *FIS* to obtain a crisp *LAG* index. Finally, the defuzzified values of the *LAG* and resilience (*R*) were measured and then inserted as inputs in the *FIS* to obtain a comprehensive defuzzified *LARG* value. The effectiveness of the proposed *LARG* framework was validated in pharmaceutical and chemical organizations. The results revealed that the *LARG* index is an effective tool for evaluating lean, agile, green, and resilience paradigms for both organizations. In conclusion, the *LARG* index provides valuable support to decision-makers in determining a business's weaknesses and strengths and guides technical managers to possible improvement actions to enhance competitiveness and sustainability.

Keywords: lean; agile; green; resilience; *LARG*; assessment; fuzzy logic



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1. Introduction

Organizations face extreme challenges due to increased pressure from environmental consciousness and regulations, ever-changing customer expectations, demand fluctuations and market volatility, and sudden internal and unplanned external disturbances. Hence, a comprehensive assessment tool of organizational capabilities to adapt to changing market conditions, seize new opportunities, and overcome challenges is needed. Lean, agile, resilience, and green (*LARG*) paradigms are recognized as the foundations of competitiveness, and the adoption of their practices leads to increased efficiency, effectiveness, performance, and competitiveness [1,2].

The lean paradigm aims to identify and eliminate non-value-added processes and activities [3,4]. Adopting lean practices enhances quality, reduces costs and waste, and increases productivity [5–7]. The agile paradigm strengthens the ability of an organization to be flexible and responsive to rapidly changing market demands and uncertainty [8–10].

The green paradigm boosts eco-friendly behavior and commitment to green thinking by adopting environmentally friendly principles and practices that could create a balance between the environment, society, and economic aspects [11–14]. Finally, resilience is the capacity of an organizational system to anticipate risks and manage them effectively through appropriate adaptation of its actions, systems, and processes to ensure that its core functions are carried out in a stable and effective relationship with the environment [15–17].

Although various organizations reported that LARG paradigms have positively influenced their capabilities and competitiveness, a comprehensive assessment of LARG paradigms is necessary to sustain improvements and growth. Consequently, this research develops a framework for assessing the adoption of LARG paradigms and evaluating the effects of their principles and practices on business performance. Furthermore, experts often use linguistic terms during the assessment of LARG paradigms. Fuzzy logic is an effective technique to deal with fuzzy or vague evaluations [18]. This research, therefore, proposes a framework for measuring the LARG index to assess the LARG paradigms under fuzzy evaluations. The proposed LARG index helps organizations understand their strengths and weaknesses; measure their lean, agility, green, and resilience capabilities; and enables them to determine necessary improvement actions to achieve their goals and objectives and enhance capabilities and competitiveness. This research, including the introduction, is structured as follows. Section 2 presents relevant studies on the lean, agile, green, and resilience paradigms. Section 3 presents the proposed framework for measuring the LARG index. Section 4 provides two case studies for illustration. Section 5 conducts index validation. Section 6 summarizes the research conclusions.

2. Literature Review

Assessing the effective adoption of the lean, agile, green, and resilience paradigms has received significant research attention. For example, Carvalho et al. [19] proposed a composite index called the “AR index” for assessing the agility and resilience of supply chains. The index was implemented in four automotive companies: one automaker and three first-tier suppliers. Azevedo et al. [20] suggested an Ecosilient index to assess the greenness and resilience of the supply chain. Azevedo et al. [21] proposed a LARG index model to assess the behavior of companies and supply chains using fuzzy logic. Fazendeiro et al. [22] proposed a framework to assess the LARG index of a supply chain in a fuzzy context. Azevedo et al. [23] proposed a LARG index to assess companies and supply chains (SCs). A case study was implemented in an automotive supply chain. Amjad et al. [24] developed a comprehensive integration framework that integrates Industry 4.0 facets supplemented by the LARG paradigms. Salleh et al. [25] proposed LARG as a mechanism to improve the competitiveness of seaports. The compatible LARG criteria in the seaport supply chain were identified and analyzed using extensive literature and expert reviews. Fifteen experts from the maritime industry were selected to assign the relative importance of each LARG paradigm by using an Analytical Hierarchy Process. Raut et al. [26] investigated the role of big data analytics as a mediator between “sustainable supply chain business performance” and lean practices, social practices, environmental practices, organizational practices, supply chain practices, financial practices, and total quality management in thirty-seven Indian manufacturing firms. Anvari [27] combined LARG and sustainable paradigms in the supply chain management of petrochemical industries. Bottani et al. [28] proposed a framework for a LARG system for a food supply chain. Çeke [29] examined the applicability of LARG approaches to a manufacturing company in Turkey. Ferraro et al. [30] proposed a conceptual framework to assess the impact of LARG practices on logistics performance. Pabarja et al. [31] conducted an assessment model for LARG in Iran’s hospital medical equipment supply chain, especially in Hamadan. The FIS was used

to evaluate LARG. Key indicators obtained from a comprehensive review of the literature and other published reports in the field of LARG were also confirmed by a focused group of experts in the medical equipment supply chain field. Ghazvinian et al. [32] proposed an integrated approach for supplier selection based on the lean, agile, resilience, green, and sustainable (LARGS) paradigm. This approach was validated using structural equation modeling and the intuitionistic fuzzy TOPSIS method.

However, this research proposes a framework for measuring the LARG index to assess the effectiveness of lean, agile, green, and resilience paradigms using fuzzy logic techniques. The framework incorporates the principles and measures of each paradigm, which enables assessing paradigms individually and concurrently. Furthermore, the framework supports decision-makers in determining the necessary actions to enhance performance. Finally, the LARG index is implemented in two industrial organizations to validate the credibility and reliability of this index.

3. Methodology

The framework for developing the LARG index is depicted in Figure 1.

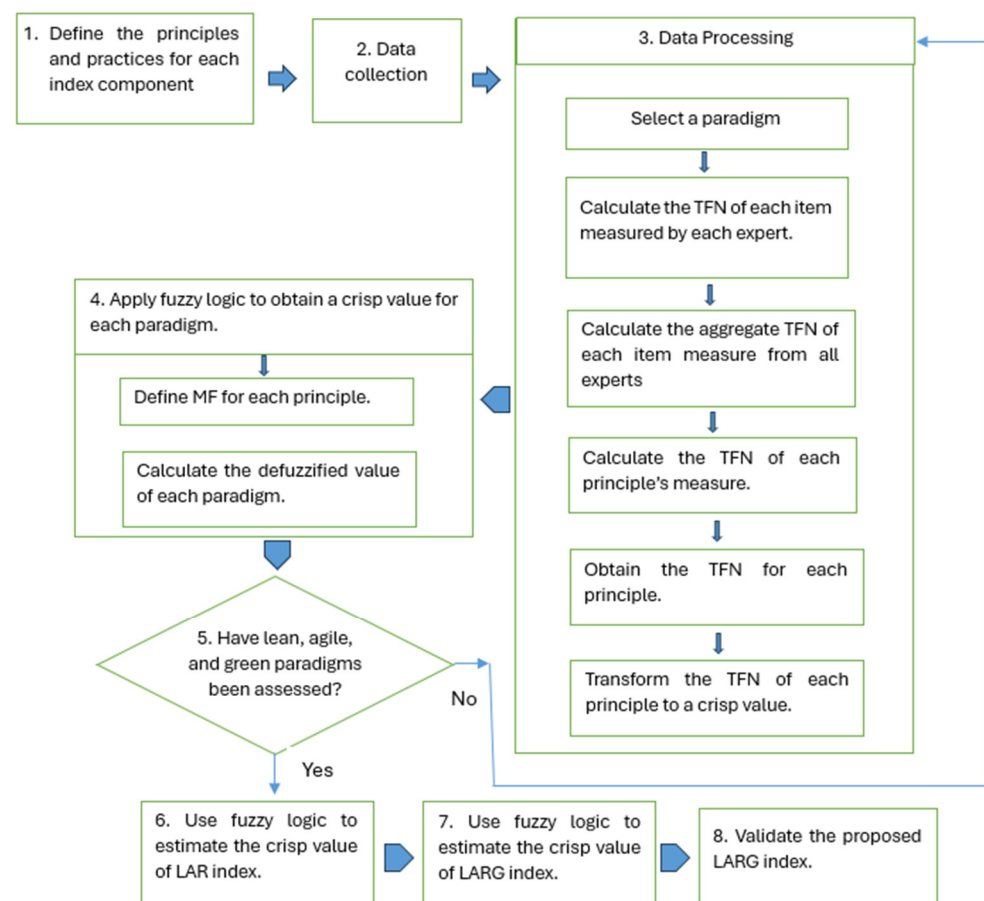


Figure 1. Proposed framework for LARG assessment.

The framework includes the following steps:

Step 1. Define the principles and practices for each index component

Table 1 lists the principles of lean, agile, and green paradigms with their corresponding measures. Four measures were identified for each paradigm. For example, the lean paradigm includes respect for people [33], value [34] and value stream [35,36], flow [37] and pull [38], and seeking perfection. Further, the agile principles cover responsiveness [39,40], concurrent engineering [41], use of information technology [42], and knowledge manage-

ment and learning [43,44]. The green principles involve reverse logistics [34,35], environmental management systems [45,46], green IT implementation [47,48], and environmental monitoring [49–51].

Table 1. Lean, agile, and green principles and measures.

Measure	Learn
Respect for People	Culture is based on respect for people. An autonomous working style. People strive for professional excellence. Promoting and motivating the ability to rapidly learn and continuously improve. Encouraging personal networks and interactions.
Value	Value is defined and understood. Processes are directed and oriented toward the quality of delivered products. The engagement of customers through product development and lifecycles.
Flow and Pull	Standardized products and parts. Optimal design of production line layout and the sequence of processes. Production smoothing/leveling and line balancing. Reducing unnecessary inventory levels. Actual production is based on customers' demands. Using Kanban cards and just-in-time tools.
Seeking Perfection	Using lessons learned to improve and become better. Using continuous improvement (Kaizen). Involvement of workers in continuous quality improvement. Involvement of customers in the continuous cycle. Monitoring production processes.
Agile	
Responsiveness	Management organization structure. Responsiveness to unpredictable demand. Ability to change delivery times of supplier's order.
Concurrent Engineering	Concurrent design of products and processes. Cross-functional and collaborative team-working environment.
Use of Information Technology	Enterprise resource planning. Computer-aided design/computer-aided manufacturing. Electronic commerce. Automated guided vehicle systems; automated storage and retrieval systems.
Knowledge Management and Learning	Strategy and leadership. Corporate culture and people. People create and share knowledge. Technology infrastructure that contributes to knowledge management. Information technology allows organizations to create, share, store, and use knowledge.
Green	
Reverse Logistics	Returned logistics. Waste logistics.
Environmental Management System	ISO 14001:2015 Eco-management and audit scheme.
Green IT Implementation	Technology advancement adoption. Spreading awareness of greenness through information technology (IT).
Environmental Monitoring	Reduce energy consumption. Reduce hazardous waste.

The resilience principles include responding, monitoring, learning, and anticipating [52]. The practices for each resilience principle are displayed in Table 2 [53–55].

Table 2. Resilience principles.

Responding	
Robustness	Production lines can withstand disturbances without function losses. The production line can function appropriately after any disturbances and restore its functionality. The damage propagation of disturbances can be prevented.
Emergency Preparedness	Availability of emergency responses to disturbances. Availability of a qualified emergency management team. Availability of emergency systems and deployment of disturbances procedures testing.
Flexibility	The ability of a production line to manage and maintain its functionality in the presence of disturbances. The ability of a production line to restructure itself.
Recoverability	Level of recovery. Recovery time; how fast the recovery occurs.
Redundancy	Availability of redundant equipment that performs the same functions when disturbances occur. Maintenance personnel should be available to substitute disturbance specialized personnel when needed.
Resource Availability	Internal and external resources are required to respond to the disturbances. Specialized maintenance technicians are required to respond to the disturbances.
Rapidity.	A failure management team that quickly responds to disturbances.
Monitoring	
Resourcefulness	Diagnosing and prioritizing disturbances by identifying and monitoring resources. Requiring initial solutions in terms of adequate material, monetary, informational, technological, and human resources when disturbances occur.
Controllability	The effectiveness of controlling and tuning disturbances.
Minimization of Disturbances and Limitation of Effects	Production lines taking preventive measures. Maintenance teams conduct frequent inspections and know the impacts of disturbances. The ability of production lines to reduce disturbances and consequences.
Learning	
Adaptability	The ability of the production line to adapt to disturbances without performance losses in the future. The availability of adjustments and improvements needed to cope with disturbances in the future.
Administration Controls and Procedures.	Training and standard operating and maintenance procedures to deal with disturbances. Training maintenance teams to increase their competence level to face disturbances.
Learning Capability.	Capability to learn from experience, knowledge, and disruptions to deal with future disturbances. An effective internal audit. Analysis and comparison assessment before and after disturbances.
Reporting Culture	A culture that reports disturbances in terms of causes and impacts to responsible and accountable personnel in organizations. Control and focus group discussions to determine the causes and effects of disturbances. Documentation of corrective preventive and safety actions taken to deal with disturbances.
Communicating Culture	Communicating and coordinating lessons learned about disturbances.
Anticipating	
Early Detection	Availability of detection procedures and instruments required for disturbances. Implementation of controls and sensors. A continuous assessment of controls and sensors.
Anticipation Capability/Forecast Accuracy and Proactivity Preparedness.	The maintenance teams extract trends and obtain warnings. The maintenance teams predict and avoid disturbances in the future. Mechanical database and maintenance logs are available to detect relevant disturbances. Planning strategies and contingency plans are required for disturbances.

Step 2. Data collection

The item measures were determined for each principle. Data were collected via a survey constructed based on an intensive review of relevant research and expert opinions [56]. The survey included 118 item measures for lean, agile, green, and resilience principles and practices. Tables 3 and 4 display the item measures for the lean principle “respect for people” and the corresponding expert evaluations, respectively. The four experts, E1–E4, were assigned weights of 0.4, 0.2, 0.2, and 0.2, respectively. The Delphi method was used for expert evaluations of item measures using linguistic terms from very low (VL) to very high (VH), as shown in Table 5.

Table 3. Likert scale.

Linguistic Term	Respect for People	Fuzzy Rating
VH	5: Very Frequently	(4, 5, 5)
H	4: Frequently	(3, 3.75, 4.5)
M	3: Occasionally	(1.75, 2.5, 3.25)
L	2: Rarely	(0.5, 1.25, 2)
VL	1: Very Rarely	(0, 0, 1)

Table 4. Item measures of the “respect for people” principle of lean.

Measure	Item Measure
A culture of respect	Q1: The organization’s culture is based on respect for people. Q2: All employees are treated the same. Q3: Diversity is welcomed, respected, and appreciated.
Autonomous working	Q4: Employees can create their schedule plans to finish assigned tasks. Q5: Employees feel responsible and accountable for their work activities.
Professional excellence and rapid learning	Q6: Top management conducts frequent training to sharpen employees’ skills. Q7: Top management strives for professional excellence. Employees are motivated to learn new techniques.
Personal networks and interaction	Q8: The organization supports group work and teamwork. Q9: Employees form operational personal networks.

Table 5. Expert responses to the lean principle “respect for people”.

Expert	Item Measures									
	Culture of Respect			Autonomous Working		Professional Excellence and Rapid Learning			Personal Networks and Interaction	
	Q1	Q2	Q3	Q3	Q4	Q5	Q6	Q7	Q8	Q9
E1	VH	VH	VH	M	H	H	VH	H	H	H
E2	H	H	VH	H	H	VH	VH	VH	H	M
E3	H	M	VH	H	M	H	VH	H	H	H
E4	VH	H	H	H	M	M	H	H	H	H

Step 3: Data processing. The responses to item measures were processed as follows:

Step 3.1: Calculate the triangular fuzzy number (TFN). For illustration, Tables 4 and 5 display the item measures for the lean principle “respect for people” and the corresponding expert evaluations, respectively. The four experts, E1–E4, were assigned weights of 0.4, 0.2, 0.2, and 0.2, respectively.

Transform the responses for each item measure (question) into a triangular fuzzy number (TFN). For illustration, the TFN of the item measure “a culture of respect” in Table 5 is calculated as follows:

Step 3.1: Calculate the TFN of each item measure by multiplying the expert weight by the corresponding fuzzy rating. For example, the rating (VH) of Q1 by expert 1 is calculated as $0.4 \times (4, 5, 5) = (1.6, 2, 2)$. The TFN of Q1 by experts 2, 3, and 4 are calculated similarly: $(0.6, 0.75, 0.9)$, $(0.6, 0.75, 0.9)$, and $(0.8, 1.0, 1.0)$, respectively.

Step 3.2: Calculate the aggregate TFN from all four experts for each question. For example, the aggregate TFN for Q1 is $(1.6 + 0.6 + 0.6 + 0.8, 2 + 0.75 + 0.75 + 1, 2 + 0.9 + 0.9 + 1) = (3.6, 4.5, 4.8)$. Similarly, the TFNs of Q2 and Q3 are calculated as 3.15 and 3.8, respectively.

Step 3.3: Calculate the TFN of the measures for each measure. For example, the TFN of the culture of respect measure $(3.52, 4.42, 4.72)$ is obtained as follows: $\left(\frac{3.6+3.15+3.8}{3}, \frac{4.5+4+4.75}{3}, \frac{4.8+4.45+4.9}{3}\right)$

Similarly, the TFNs of autonomous working, professional excellence and rapid learning, and personal networks and interaction are calculated as $(2.63, 3.38, 4.13)$, $(3.31, 4.17, 4.62)$, and $(2.88, 3.63, 4.38)$, respectively.

Step 3.4: Obtain the TFN of each principle by dividing the sum of TFNs of its measures by the number of measures. For example, the TFN of “respect for people” is calculated from its four measures as follows:

$$= \left(\frac{3.52 + 2.63 + 3.32 + 2.875}{4}, \frac{4.42 + 3.38 + 4.17 + 3.63}{4}, \frac{4.72 + 4.13 + 4.62 + 4.38}{4} \right) = (3.08, 3.90, 4.46)$$

Step 3.5: Transform the TFN of each principle to a crisp value, D , using Equation (1), where U , M , and L are the upper, middle, and lower elements of the TFN of the principle.

$$D = \frac{1}{3}((U - L) + (M - L)) + L \quad (1)$$

For example, the D value for the “respect for people” lean principle is calculated as:

$$D = \frac{1}{3}((4.46 - 3.08) + (3.90 - 3.08)) + 3.90 = 3.81$$

Similarly, the calculated D values for value, flow and pull, and seeking perfection principles of lean are 3.66, 2.87, and 3.59, respectively.

Step 4: Fuzzy logic theory can deal with imprecise, uncertain, vague, and subjective judgment problems [57]. The fuzzy logic system consists of three main steps: fuzzification, fuzzy inference system (FIS), and defuzzification. Apply fuzzy logic to calculate the defuzzified values for each principle as follows:

Step 4.1: Define membership functions (MF) for each principle, as shown in Figure 2, utilizing the fuzzy ratings (VL to VH) in Table 5. Then, establish the If–Then rules that relate inputs of the FIS to outputs, as shown in Figure 3.

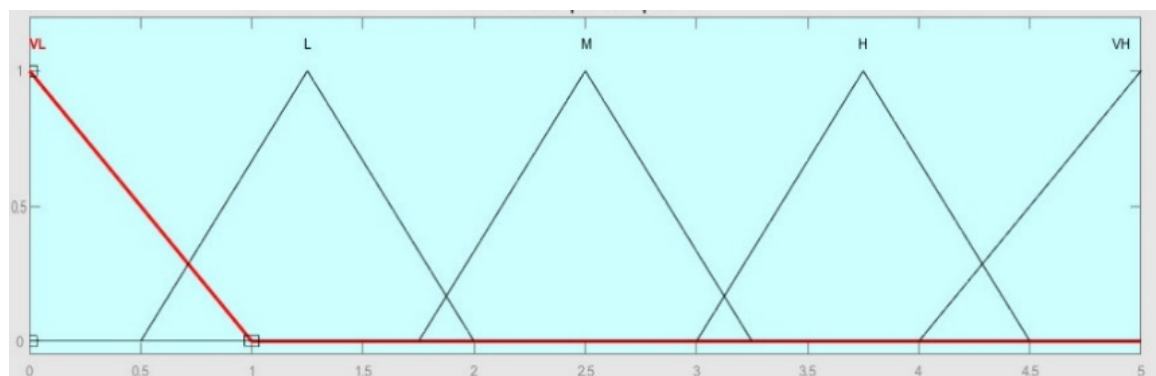


Figure 2. The MFs for lean, agile, green, and resilience.

Figure 3. A sample of the If–Then rules for resilience.

For illustration, a sample of the If–Then rules that relate the four measures: respect for people (I_1), value (I_2), flow and pull (I_3), and seeking perfection (I_4) principles for lean (L) as an output in the FIS are shown in Table 6.

Table 6. Sample of the If–Then rules for lean. (I_1 is VL and I_2 is VL).

		I_3				
		VL	L	M	H	VH
I_4	VL	VL	VL	VL	L	L
	L	VL	VL	L	L	L
	M	VL	L	L	M	M
	H	VL	L	M	M	M
	VH	L	L	M	M	M

Step 4.2. Calculate the defuzzified value of each paradigm using the Mamdani inference system. The output of each rule is a fuzzy set. The output fuzzy sets are then combined into a single fuzzy set using the aggregation method (Max) of the FIS. Then, defuzzification is employed to obtain a crisp output value using centroid defuzzification. Figure 4 illustrates the use of fuzzy logic for the lean paradigm.

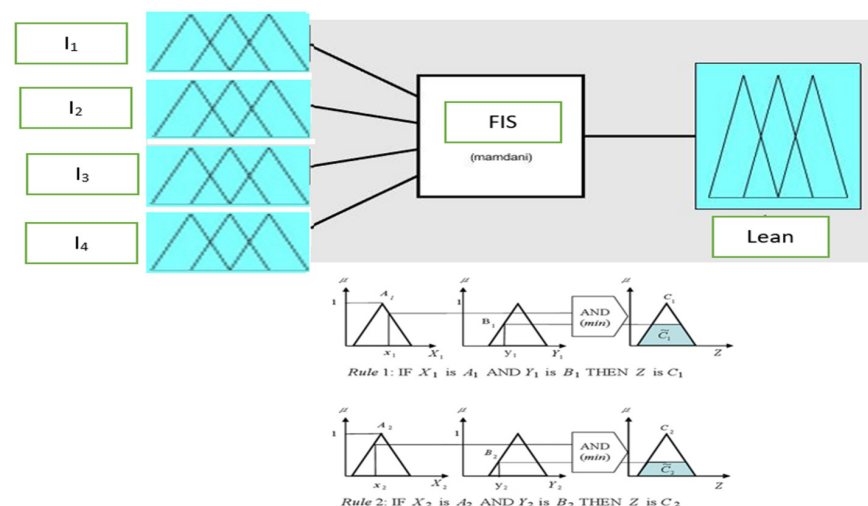


Figure 4. Fuzzy logic for lean.

Step 5. Repeat steps 1 to 4 to calculate the crisp values for agility, green, and resilience.

Step 6. Use fuzzy logic to calculate the crisp values of the LAG index, where the inputs of the *FIS* are lean, agile, and green, while the output is the LAG index, as shown in Figure 5.

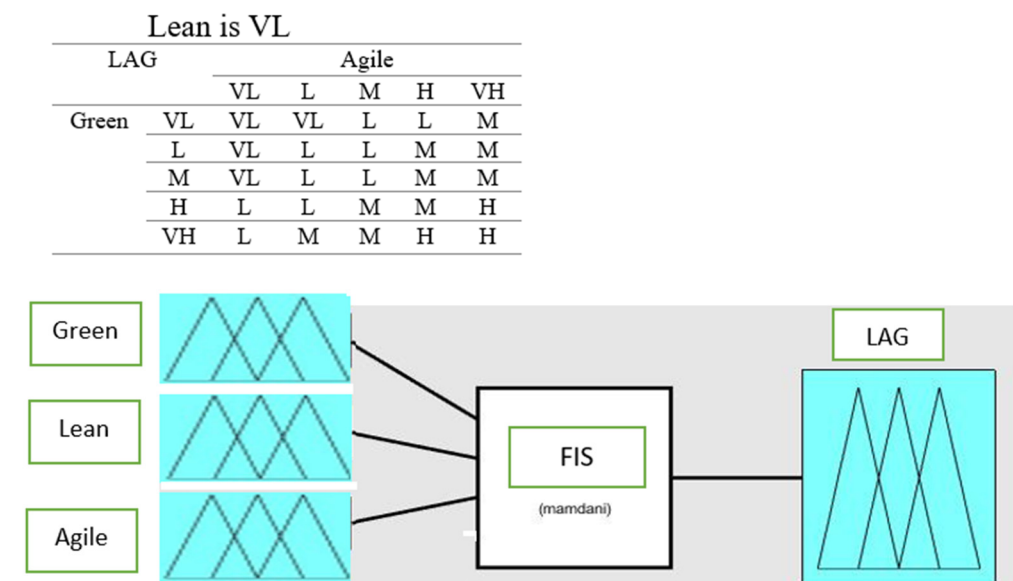


Figure 5. Fuzzy logic for LAG.

Step 7. Use fuzzy logic to estimate the crisp value of the LARG index. The inputs of the *FIS* are crisp values of the LAG index and resilience, while the output is the LARG index, as shown in Figure 6.

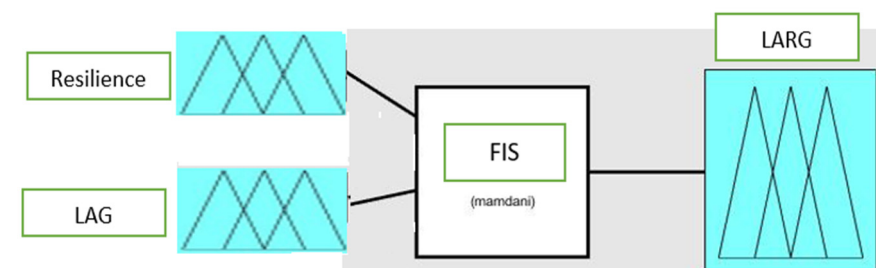


Figure 6. Fuzzy logic for LARG.

Step 8. Validate the proposed LARG index in real-life applications. Discuss the results obtained and then suggest recommendations.

4. Applications of the LARG Index

Two firms specialized in chemicals (C) and pharmaceutical (P) products were considered to validate the reliability of the LARG index. Four experts, who have solid knowledge and experience in lean, agile, green, and resilience principles and practices, evaluated the survey. Then, *TFNs* of measures were calculated for all principles. Finally, *TFNs* of each principle were estimated for lean, agile, green, and resilience paradigms. Tables 7–10 display the *TFNs* for the principles and measures of *LARG* paradigms, respectively.

Table 7. The TFNs of resilience principles and measures.

Principle	Measure	TFN of Measure			TFN of Principle	
					Chemicals	Pharmaceutical
Responding	Robustness	2.87	3.67	4.27	(2.9, 3.70, 4.29)	(2.67, 3.48, 4.06)
	Emergency preparedness.	3.18	4.00	4.53		
	Flexibility.	2.13	2.88	3.63		
	Recoverability	2.83	3.63	4.23		
	Redundancy	3.00	3.75	4.50		
	Resource availability	2.70	3.50	4.10		
Monitoring	Rapidity.	3.60	4.50	4.80	(3.60, 3.47, 4.22)	(2.59, 3.42, 3.95)
	Resourcefulness.	2.33	3.08	3.83		
	Controllability.	3.00	3.75	4.50		
	Minimization of disturbances and limitation of effects.	2.83	3.58	4.33		
Learning	Adaptability	3.10	4.00	4.30	(2.62, 3.40, 4.06)	(3.25, 4.10, 4.55)
	Administration controls and procedures.	2.75	3.50	4.25		
	Learning capability.	2.42	3.17	3.92		
	Reporting culture	2.58	3.33	4.08		
	Communicating culture	2.25	3.00	3.75		
Anticipating	Early detection	2.83	3.58	4.33	(2.51, 3.26, 4.01)	(2.85, 3.65, 4.26)
	Anticipation capability/forecast accuracy and proactivity preparedness.	2.19	2.94	3.69		

Table 8. The TFNs of agility principles and measures.

Principle	Measure	TFN of Measure			TFN of Principle	
					Chemicals	Pharmaceutical
Responsive supply chain	Organizational structure	2.58	3.33	4.08	(2.36, 3.11, 3.86)	(2.85, 3.65, 4.26)
	Responsiveness to unpredictable demand	1.75	2.5	3.25		
	Ability to change delivery times of supplier's order	2.75	3.5	4.25		
Concurrent engineering	Concurrent design of products and processes	2.625	3.375	4.125	(2.63, 3.38, 4.13)	(3.27, 4.08, 4.63)
	Multidisciplinary team working environment	2.25	3	3.75		
	Collaborative work	3	3.75	4.5		
Use of information technology	ERP	2.375	3.125	3.875	(2.22, 2.97, 3.72)	(2.79, 3.59, 4.19)
	CAD/CAM	2.25	3	3.75		
	Electronic commerce	2.5	3.25	4		
	AGVs AS AR	1.75	2.5	3.25		
Knowledge management and learning	Strategy and leadership	3	3.75	4.5	(2.47, 3.22, 3.97)	(3.23, 4.06, 4.57)
	Corporate culture and people	1.92	2.67	3.42		
	Technology	2.5	3.25	4		

Table 9. The TFNs of lean principles and measures.

Principle	Measure	TFN of Measure			TFN of Principle/Fuzzified Value	
					Chemicals	Pharmaceutical
Respect for people	Culture is based on respect for people	3.0	3.75	4.1	(3.03, 3.84, 4.31)	(3.08, 3.90, 4.46)
	An autonomous working style	2.63	3.38	4.13		
	Striving for professional excellence and promoting rapid learning	2.95	3.75	4.35		
	Encouraging personal networks and interaction	2.95	3.75	4.35		
Value and value stream	Value is defined and understood	1.85	2.38	3.2	(2.68, 3.44, 4.05)	(2.93, 3.75, 4.30)
	All processing on products is directed towards the benefits the product deliver	2.8	3.63	4.15		
	Engagement of customers	2.75	3.5	4.25		
	Defining and understanding all activities	3.6	4.5	4.8		
	Application of value stream mapping	2.925	3.75	4.28		
	Eliminating non-value-added processes and activities	2.13	2.88	3.63		
Flow and pull	Standardized products	2.3	3.13	3.65	(2.43, 3.17, 3.85)	(2.17, 2.88, 3.56)
	Optimal design	3.05	3.88	4.4		
	Production smoothing	2.5	3.25	4		
	Customer demand	2.625	3.38	4.13		
	Level of inventory	3	3.75	4.5		
	Kanban cards	1.1	1.63	2.45		
Perfection	Continuous improvement	2.18	3	3.53	(2.18, 3.00, 3.53)	(2.85, 3.63, 4.30)
	Involvement of workers and customers	3.35	4.25	4.55		
	Monitoring production	3.3	4.125	4.65		

Table 10. The TFNs of green principles and measures.

Principle	Measure	TFN of Measure			TFN of Principle/Fuzzified Value	
					Chemicals	Pharmaceutical
Reverse logistics	Returned logistics	2.5	3.25	4.0	(2.0, 2.75, 3.5)	(1.76, 2.46, 3.16)
	Waste logistics	1.5	2.25	3.0		
	ISO	1.875	2.625	3.375	(1.81, 2.56, 3.31)	(2.31, 3.06, 3.81)
	EU	1.75	2.5	3.25		
Green IT implementation	Technology advancement	1.67	2.42	3.17	(1.46, 2.21, 2.96)	(2.23, 2.96, 3.96)
	Awareness of green	1.25	2.00	2.75		
Environmental monitoring of suppliers	Eco-friendly	2.5	3.25	4.0	(2.0, 2.75, 3.50)	(2.05, 2.75, 3.38)
	Reduces energy	1.75	2.5	3.25		
	Reduces hazard	1.75	2.5	3.25		

The *TFNs* of principle measures were transformed into crisp values. Fuzzy logic was then utilized to determine the defuzzified values of LARG paradigms from their principles. Table 11 displays the resulting defuzzified values for both industries.

Table 11. The defuzzified values of principles and measures.

Paradigm	Principle	Chemicals	Pharmaceutical
Lean		3.75	3.75
	L ₁ : Respect for people	3.73	3.81
	L ₂ : Value and value stream	3.39	3.66
	L ₃ : Flow and pull	3.15	2.87
	L ₄ : Perfection	3.66	3.59
Agile		3.07	3.75
	A ₁ : Responsive supply chain	3.11	3.16
	A ₂ : Concurrent engineering	3.38	3.99
	A ₃ : Use of information technology	2.97	3.53
	A ₄ : Knowledge management and learning	3.22	3.95
Green		2.5	2.5
	G ₁ : Reverse logistics	2.75	2.46
	G ₂ : Green IT implementation	2.56	3.06
	G ₃ : Environmental monitoring of suppliers	2.75	2.73
Resilience		3.75	3.75
	R ₁ : Responding	3.63	3.4
	R ₂ : Monitoring	3.47	3.32
	R ₃ : Learning	3.36	3.97
	R ₄ : Anticipating	3.26	3.58
LAG		2.87	3.75
LAG-R		2.50	3.75

From Table 11, the following results are obtained:

- For the lean paradigm, the highest scores of 3.73 and 3.81 correspond to the respect for people principle (L₁) in both the chemical (C) and pharmaceutical (P) industries. However, the lowest scores of 3.15 and 2.87 correspond to flow and pull (L₃).
- For the agility paradigm, concurrent engineering (A₂) achieved the highest scores of 3.38 and 3.99 in both the C and P industries. However, the lowest scores of 2.97 and 3.16 correspond to the use of information technology (A₃) and responsive supply chain (A₁), respectively.
- For the green paradigm, the highest scores of 2.75 and 3.06 in the C and P industries correspond to reverse logistics (G₁), environmental monitoring of suppliers (G₃), and green IT implementation (G₂), respectively. However, their lowest scores of 2.56 and 2.46 correspond to G₂ and G₁, respectively.
- For the resilience paradigm, the highest scores of 3.63 and 3.97 correspond to the responding (R₁) and learning (R₃) principles in the C and P industries, respectively. However, their lowest scores correspond to anticipating (R₄) and monitoring (R₂), 3.26 and 3.32, respectively.

5. Index Validation

Figure 7 conducts a comparison of the average score of paradigm measures between the C and P industries, where the following is noted:

- The pharmaceutical company has higher scores in L₁ and L₂. However, the chemical company surpasses the pharmaceutical company in L₃ and L₄.
- The G₂ score in the pharmaceutical company is significantly higher than in the chemical company.
- The pharmaceutical company shows higher agility than the chemical company in all principles.
- The chemical company has higher scores in R₁ and R₂. However, the pharmaceutical company surpasses the pharmaceutical company in R₃ and R₄.

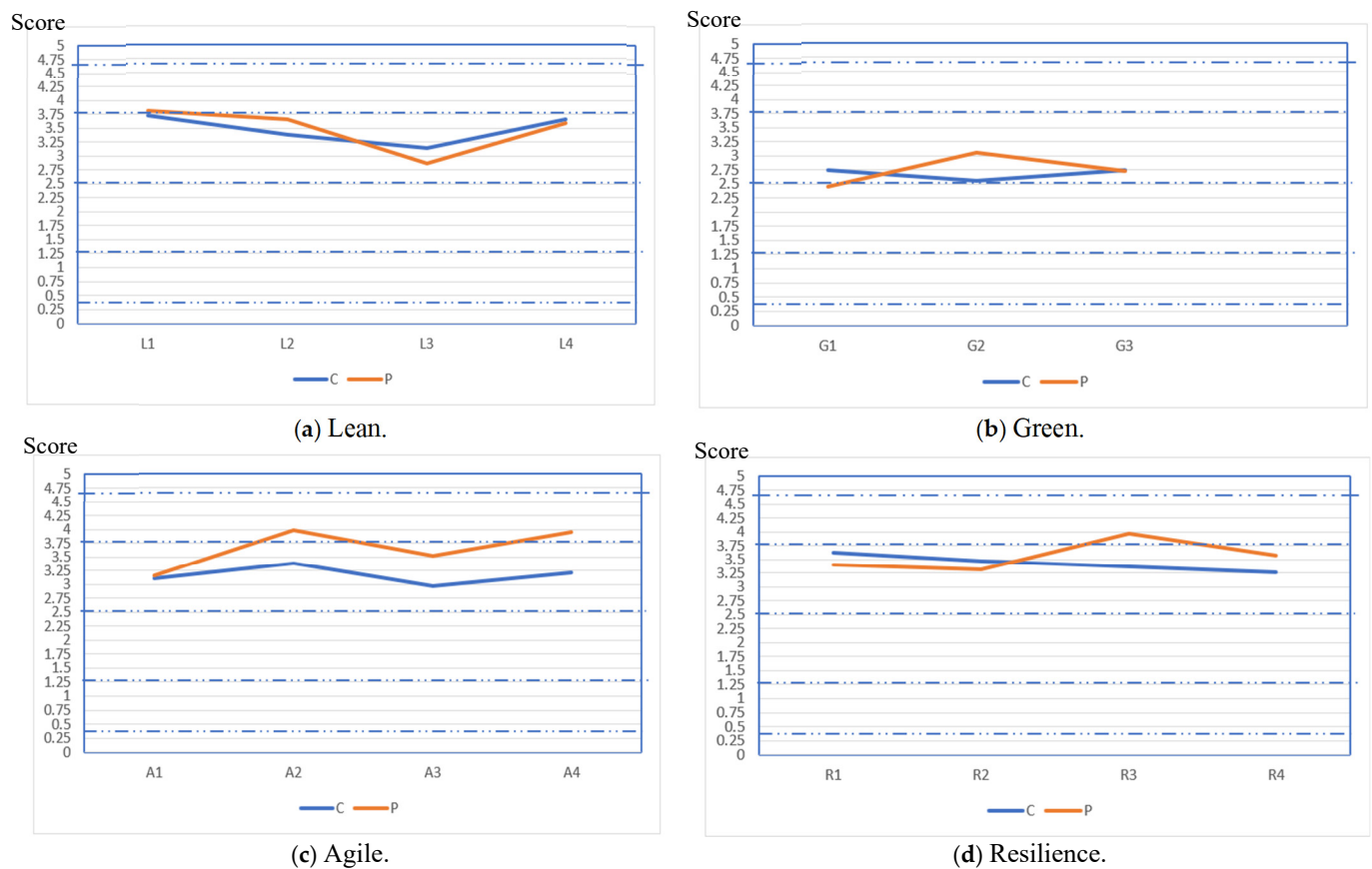


Figure 7. A comparison of the average scores of paradigm measures between the C and P industries.

Table 12 displays the rating to assess the principles and paradigms. Table 13 displays the assessment results of the principles and paradigms in both industries, where the following is found:

- The level of lean adoption is high (H) in both industries. Meanwhile, opportunities for improvement can be made to enhance the “flow and pull” principle in both industries.
- The agility is moderately high (M-H) and high in the C and P industries, respectively. The chemical industry should enhance the principles of “responsive supply chain”, “use of information technology”, and “knowledge management and learning”. Further, the pharmaceutical industry should enhance the principle “responsive supply chain”.
- The overall level of the green paradigm is moderate (M) in both industries. Hence, enhancement of the three principles of “reverse logistics”, “green IT implementation”, and “Environmental monitoring of suppliers” are needed in both industries.
- The resilience level is high in both industries. Moreover, high levels of responding, learning, anticipating, and monitoring are achieved in both firms.
- Combining the lean, agile, and green paradigms, the LAG values are moderate (=2.87) and high (=3.75) in the C and P industries, respectively. This proves that the LAG index effectively assesses the overall levels of lean, agile, and green in both firms and can provide a reliable overall assessment.
- Considering the lean, agile, green, and resilience paradigms, the levels of the LARG index are moderate (2.5) and high (3.75) in C and P, respectively. This proves the reliability and effectiveness of the LARG index in assessing the overall level of the four paradigms.

Table 12. Assessment ranges.

Linguistic Term	Range
VH: Very High	4.5–5
H-VH: High–Very High	>4–<4.5
H: High	>3.25–<4
M-H: Moderate–High	>3–<3.25
M: Moderate	>2–3
L-M: Low–Moderate	>1.75–2
L: Low	>1.00–1.75
VL-L: Very Low–Low	>0.5–1.00
VL: Very Low	0–0.5

Table 13. Assessment of lean, agile, green, and resilience paradigms.

Paradigm	Principle	C	P
Lean		H	H
	L ₁ : Respect for people	H	H
	L ₂ : Value and value stream	H	H
	L ₃ : Flow and pull	M-H	M
	L ₄ : Perfection	H	H
Agile		M-H	H
	A ₁ : Responsive supply chain	M-H	M-H
	A ₂ : Concurrent engineering	H	H
	A ₃ : Use of information technology	M	H
	A ₄ : Knowledge management and learning	M-H	H
Green		M	M
	G ₁ : Reverse logistics	M	M
	G ₂ : Green IT implementation	M	M-H
	G ₃ : Environmental monitoring of suppliers	M	M
Resilience		H	H
	R ₁ : Responding	H	H
	R ₂ : Monitoring	H	H
	R ₃ : Learning	H	H
	R ₄ : Anticipating	H	H
LAG		M	H
LARG		M	H

Compared to previous studies, the proposed index can provide the following advantages:

- This study provides a comprehensive assessment of the lean, agile, green, and resilience paradigms by assessing their principles and measures. This enables decision-makers to identify weaknesses and strengths in their systems by assessing the measures of each principle.
- The LARG provides a reliable overall assessment of lean, agile, green, and resilience paradigms, as validated in two distinct industries.
- The assessment requires moderate skills in fuzzy logic, which makes it easily understandable and applicable to a wide range of business applications.
- The proposed framework provides an effective assessment of each paradigm separately. This supports managers in prioritizing their efforts on the important paradigms.

6. Conclusions

The effective adoption of *LARG* paradigms has significantly enhanced organizations' competitive advantages. To provide a quantitative assessment of *LARG* progress, this research developed a framework for assessing the effectiveness of *LARG* paradigms using the *LARG* index. The principles and their measures of *LARG* paradigms were defined. Then, a questionnaire was prepared and filled with four experts based on a five-point Likert scale. Using fuzzy logic, the defuzzified value was obtained for each of the *LARG* paradigms. Two case studies, pharmaceutical and chemical companies, were employed to illustrate and validate the proposed *LARG* index, in both of which the *LARG* index was found to be effective in evaluating the progress of each paradigm as well as of combined paradigms. Companies can utilize this index to assess the capabilities of their systems regarding lean, agile, green, and resilience paradigms. Moreover, the detailed evaluation of the *LARG* index enables organizations to determine the strengths and weaknesses in each paradigm and support them in planning and directing improvement actions. In conclusion, organizations can utilize the proposed *LARG* index to assess their improvement strategies and identify improvement directions and actions to enhance their operational, economic, and environmental performance. This research assumes that paradigms are independent; hence, each paradigm was assessed separately. Future research should consider developing an interpretive structural model for deploying lean, agile, green, and resilience principles enabled by Industry 4.0.

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